

Name: Dylan Shaffer
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Module: M3

In this homework, you will do a Naive Bayes Classification by hand. Our data for the M3 homework is a modified version of a list of Pokemon of Water, Normal, and Flying type. We will be using a smaller version with 12 example Pokemon.

Name	HP	Attack	Defense	Type
Vigoro	Mid	High	Mid	Normal
Dodrio	High	Mid	High	Normal
Farfetch'd	Low	High	Low	Normal
Lillipup	Low	Low	Low	Normal
Jellicent	Mid	Low	High	Water
Corsola	Low	Low	Mid	Water
Piplup	Low	Low	Low	Water
Marill	High	Low	Low	Water
Landorus Therian Forme	Mid	Mid	Mid	Flying
Beautifly	High	High	Low	Flying
Tropius	Mid	High	Mid	Flying
Drifblim	Mid	High	Low	Flying

Our overall goal is to look at a Pokemon outside of our dataset to see whether the Type is more likely Normal, Water, or Flying. In particular, we will estimate this for the *Mandibuzz* Pokemon, which has **HP=Mid, Attack=High, and Defense=Mid**.

Question 1:

Calculate the Prior Probability of $P(y)$ for Type across the whole dataset:

Answer 1:

$$p(\text{Type}=\text{Normal}) = 0.33333333$$

$$p(\text{Type}=\text{Water}) = 0.33333333$$

$$p(\text{Type}=\text{Flying}) = 0.33333333$$

Question 2:

Please calculate a subset of the likelihood for each of the features HP, Attack, and Defense for each feature.

Answer 2:**For HP:**

$$p(\text{HP}=\text{Mid} \mid \text{Type}=\text{Normal}) = 0.25$$

$$p(\text{HP}=\text{Mid} \mid \text{Type}=\text{Water}) = 0.25$$

$$p(\text{HP}=\text{Mid} \mid \text{Type}=\text{Flying}) = 0.75$$

For Attack:

$$p(\text{Attack}=\text{High} \mid \text{Type}=\text{Normal}) = 0.5$$

$$p(\text{Attack}=\text{High} \mid \text{Type}=\text{Water}) = 0$$

$$p(\text{Attack}=\text{High} \mid \text{Type}=\text{Flying}) = 0.75$$

For Defense:

$$p(\text{Defense}=\text{Mid} \mid \text{Type}=\text{Normal}) = 0.25$$

$$p(\text{Defense}=\text{Mid} \mid \text{Type}=\text{Water}) = 0.25$$

$$p(\text{Defense}=\text{Mid} \mid \text{Type}=\text{Flying}) = 0.5$$

Question 3:

Calculate the posterior probability for each Type, given the features of *Mandibuzz* which we will represent as “Features”: (**HP=Mid, Attack=High, and Defense=Mid**). Using the assumption that each feature (HP, Attack, and Defense) is independent, then this can be calculated by applying bayes law like the following:

For example,

$p(\text{Type}=\text{Normal} \mid \text{Features}) =$

$$\frac{p(\text{HP} = \text{Mid} \mid \text{Type} = \text{Normal}) * p(\text{Attack} = \text{High} \mid \text{Type} = \text{Normal}) * p(\text{Defense} = \text{Mid} \mid \text{Type} = \text{Normal}) * p(\text{Type} = \text{Normal})}{p(\text{Features})}$$

Here, you can calculate $p(\text{Features})$ =:

$$\begin{aligned} & p(\text{Features} \mid \text{Type} = \text{Normal}) * p(\text{Normal}) + \\ & p(\text{Features} \mid \text{Type} = \text{Water}) * p(\text{Type} = \text{Water}) + \\ & p(\text{Features} \mid \text{Type} = \text{Flying}) * p(\text{Type} = \text{Flying}) \end{aligned}$$

Answer 3:

$$p(\text{Type}=\text{Normal} \mid \text{Features}) = 0.008928571$$

$$p(\text{Type}=\text{Water} \mid \text{Features}) = 0$$

$$p(\text{Type}=\text{Flying} \mid \text{Features}) = 0.080357143$$

Question 4:

Which Pokeymon Type (i.e. Normal, Water, Mid) has the highest probability and should be our predicted answer?

Answer 4:

The highest probability, thus predicted answer, is Flying type.

